

**Small-Object Detection in Water Bodies Using the Data
Captured by USV**

A PROJECT REPORT

Submitted by

BL.EN.U4CSE20067	Revanth Kandimalla
BL.EN.U4CSE20076	Sumanth K
BL.EN.U4CSE20103	Mourya Vardhan Reddy
BL.EN.U4CSE20187	Nishitha U

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING



AMRITA SCHOOL OF COMPUTING, BENGALURU

AMRITA VISHWA VIDYAPEETHAM

BENGALURU 560 035

ACKNOWLEDGEMENTS

The satisfaction that accompanies successful completion of any task would be incomplete without mention of people who made it possible, and whose constant encouragement and guidance have been source of inspiration throughout the course of this project work.

We offer our sincere pranams at the lotus feet of **"AMMA", MATA AMRITANANDAMAYI DEVI** who showered her blessing upon us throughout the course of this project work.

We owe our gratitude to **Prof. Manoj P.**, Director, Amrita Vishwa Vidyapeetham Bengaluru Campus. We thank **Dr. Sriram Devanathan**, Principal Amrita School of Computing, Bengaluru for his support and inspiration.

We would like place our heartfelt gratitude to **Dr. Gopalakrishnan E.A.**, Chair, Department of Computer Science and Engineering, Amrita School of Computing, Bengaluru for his valuable support and inspiration.

It is a great pleasure to express our gratitude and indebtedness to our project guide Dr. Rimjhim, Assistant Professor, Department of Computer Science and Engineering, Amrita School of Computing, Bengaluru for her/his valuable guidance, encouragement, moral support, and affection throughout the project work.

We would like to thank express our gratitude to project panel members for their suggestions, encouragement, and moral support during the process of project work and all faculty members for their academic support. Finally, we are forever grateful to our parents, who have loved, supported and encouraged us in all our endeavors.

ABSTRACT

The contamination of natural water bodies with small plastic water bottles and cans poses a pressing environmental challenge. To address this issue, this project focuses on the development of an automated system for small plastic water bottles and cans detection within water bodies. In this project, we lay the foundation for this system by leveraging Unmanned Surface Vehicles (USVs) to capture high-resolution imagery of water bodies. The primary objective is to demonstrate the feasibility of using computer vision technology to detect and quantify the presence of small plastic bottles and cans in water bodies. Convolutional Neural Networks (CNNs) are highly effective for object detection tasks. Techniques like Faster R-CNN, YOLO (You Only Look Once), and SSD (Single Shot MultiBox Detector) have shown excellent performance in detecting objects within images. These networks can be trained on a labeled dataset of water bottles and cans to identify them accurately. Apart from Deep learning techniques, Image segmentation techniques can be used to separate objects of interest (water bottles and cans) from the background. In this project we will be working on Semantic Segmentation and Instance Segmentation. Semantic Segmentation is a technique which assigns each pixel in the image to a class (e.g., "water bottle," "can," or "background"), creating a pixel-wise mask of the objects. And Instance Segmentation goes a step further and not only classifies objects but also distinguishes between individual instances of the same class. It can be useful when you want to count the number of bottles or cans.

TABLE OF CONTENTS

1. Page No.

ACKNOWLEDGEMENTS	i
ABSTRACT	ii
TABLE OF CONTENTS	iii
LIST OF FIGURES	iv
LIST OF TABLES	iv
CHAPTER 1- INTRODUCTION	1
1.1 PROBLEM STATEMENT	1
1.2 MOTIVATION	1
1.3 OBJECTIVE	1
CHAPTER 2 – LITERATURE SURVEY	4
CHAPTER 3 – SYSTEM SPECIFICATIONS	7
3.1 SOFTWARE REQUIREMENTS	7
3.2 HARDWARE REQUIREMENTS	7
CHAPTER 4 – SYSTEM DESIGN	8
4.1 LOW LEVEL DESIGN WITH DESCRIPTION	8
4.2 HIGH LEVEL DESIGN WITH DESCRIPTION	9
CHAPTER 5 – SYSTEM IMPLEMENTATION	11
5.1 DATA PREPROCESSING	11
5.2 TRAINING	12
5.3 METRICS	13
5.4 MODULES	13
CHAPTER 6 – SYSTEM TESTING	14
CHAPTER 7 – RESULTS AND ANALYSIS	15
CHAPTER 8 - CONCLUSION AND FUTURE SCOPE	18

LIST OF FIGURES

FIGURE	PAGE.NO
Fig 1. Sample image of waste floating on water	2
Fig 2. Detection of waste floating on water	3
Fig 3. Dataset images with challenges	8,9
Fig 4. The architecture of yolov7 model	10
Fig 5. Flow chart showing the high-level system design	12
Fig 6. Proposed model architecture after fine tuning the model.	12
Fig 7. ECANET architecture	13
Fig 8 . HORNET architecture	13
Fig 9 . Performance metrics of the proposed model.	19
Fig 10. Precision Curve of the proposed model	20
Fig 11. Recall curve of the proposed model.	21
Fig 12. Precision and Recall curve of the proposed model.	21
Fig 13. F1 Score curve of the proposed model.	22
Fig 14. Confusion matrix of the proposed model.	22
Fig 15. Prediction of test images made by the proposed model	28

LIST OF TABLES

TABLE	PAGE.NO
Table 1: Hyperparameter used in model training	17
Table 2: Augmentation used in model training	18
Table 3. Results after training with different models.	24
Table 4. Results after training with different yolov7 models.	25
Table 5. Impact of Different Activation Functions on Model Performance	30

CHAPTER - 1

INTRODUCTION

1.1 PROBLEM STATEMENT

Waste in water bodies is a modern problem that has grown to alarming proportions. It is a complex problem with wide-ranging environmental effects. Globally, water bodies—from lakes and rivers to the oceans—bear the brunt of human activity that fuels the unrelenting buildup of waste. Humans are dumping waste like bottles in water bodies directly as shown in Fig 1. Aside from degrading aquatic ecosystems' aesthetic appeal, plastics, chemicals, and other non-biodegradable materials also seriously endanger aquatic life and human health. Waste is ubiquitous in water bodies, upsetting delicate ecological balances and eventually causing the deterioration of entire ecosystems and the quality of the water to decline. In order to reduce the number of pollutants entering our essential water supplies, creative solutions and increased awareness are urgently needed, as shown by the current efforts to address this issue. It is becoming more and more clear that sustainable practises and technologies that support efficient waste management in water bodies are necessary as communities deal with the fallout from careless waste disposal.

1.2 MOTIVATION

The pressing need for innovative technology to tackle environmental deterioration and pollution has been highlighted in light of the growing concerns around these issues in recent times. It is a difficult but necessary duty to detect and monitor contaminants in water bodies, especially tiny items like waste products. Modern computer vision models have enabled Unmanned Aerial Vehicles (UAVs) to become extremely effective instruments for environmental surveillance. This study uses data from unmanned aerial vehicles (UAVs) to identify tiny things in water bodies using YOLO (You Only Look Once) object detection algorithms, especially versions V5, V7, and V8.

1.3 OBJECTIVE

The YOLO framework, which is well-known for its ability to recognize objects in real time, has undergone several iterations of improvement that have increased the model's precision and effectiveness. This study investigates the ability of YOLO versions V5, V7, and V8 to identify and categorize tiny items in water bodies, with a major focus on garbage. The data collected by UAVs is an invaluable asset that provides an aerial perspective that enables thorough observation of aquatic habitats.

A crucial component of any detection system, the accuracy metrics are essential for assessing how well the models that have been put into practice are working. Our research yields encouraging findings: YOLO V5 detects tiny things in water bodies as depicted in Fig 2, with an accuracy of 34%, V7 with a 76% accuracy, and V8 with a remarkable 77.4% accuracy. These results highlight the YOLO versions' developing capabilities as well as the potential of unmanned aerial vehicles (UAVs) to transform environmental monitoring and lessen water pollution.

A crucial component of any detection system, the accuracy metrics are essential for assessing how well the models that have been put into practice are working. Our research yields encouraging findings: YOLO V5 detects tiny things in water bodies with an accuracy of 34%, V7 with a 76% accuracy, and V8 with a remarkable 77.4% accuracy. These results highlight the YOLO versions' developing capabilities as well as the potential of unmanned aerial vehicles (UAVs) to transform environmental monitoring and lessen water pollution.



Fig 1. Sample image of waste floating on water

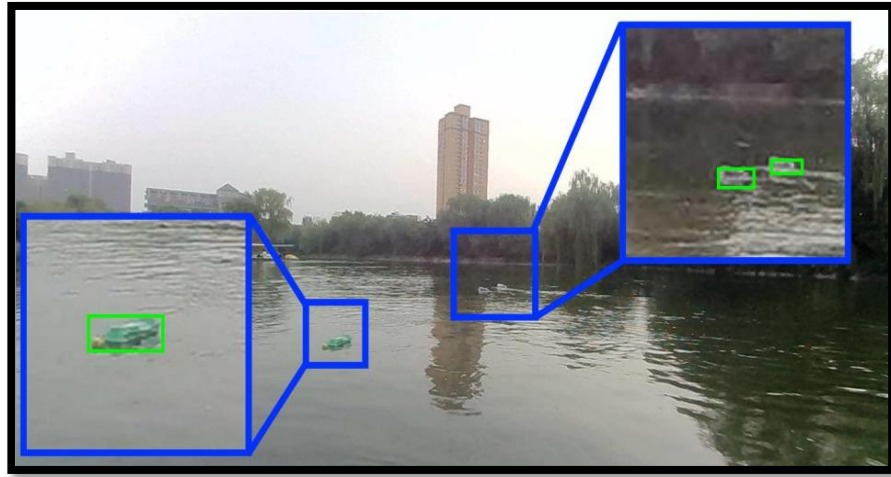


Fig 2. Detection of waste floating on water

CHAPTER - 2

LITERATURE REVIEW

Wastage dumping in water bodies has been a major issue for many decades despite the results of its effects on aquatic life and the environment. Non-biodegradable materials such as plastics, chemicals, etc are causing a major problem for aquatic ecosystems. One such way to prevent these wastes from reaching the ocean is by taking action beforehand and eliminating these wastes in inland waters. As discussed in [10], one of the solutions for the problem is to deploy multiple cleaning units along the course of the flow of water, and in inland water bodies like lakes, where water is continuously monitored and cleaned, hence preventing the debris from reaching the oceans. In this study, we assume waste to be water bottles and cans which need to be detected in order to eliminate from water bodies before they reach and contaminate the oceans in a similar method approached by [1]. The sections that follow provide a concise literature review of the literature on the use of computer vision and deep learning algorithms for water surface waste detection, as well as an overview of the development of these techniques throughout time.

One of the early publications related to waste detection using deep learning was done in 2020 [3], which was, AquaVision: Automating the detection of waste in water bodies using deep transfer learning. CNNs have emerged as the backbone of many waste detection systems within AquaVision. These models, pre-trained on extensive datasets, demonstrate an exceptional ability to learn hierarchical features crucial for identifying waste patterns in diverse water bodies. Utilizing pre-trained architectures, like ResNet, VGG, and MobileNet, demonstrates a tendency towards utilizing knowledge already obtained from extensive image datasets. Later in the year 2021, we can see many publications related to small object detection in water bodies, where different approaches were used in all these 4 publications. In 2021 [4], The use of wireless sensor signals to detect floating waste debris in water bodies has become an innovative and promising approach for environmental monitoring. The literature in this area reflects the diverse range of sensors, signal processing techniques, and communication technologies employed to create effective waste detection systems with wireless sensor signals to detect the waste debris that is floating on the water bodies. It is a different approach where the sensor detects the waste based on the change in the frequency around the waste debris. In the same year [5], a model was proposed where they used satellite images to detect the objects floating in the water bodies. The "MacroFloat," or high-resolution multispectral images, were taken by the Sentinel-2 satellite

constellation and are the dataset utilised in this study. The capacity of these photos to record fine details in the visible and infrared spectrum allows for a thorough examination of the coastal areas. The dataset used was real-time and the accuracy was pretty decent. Research on marine pollution monitoring, ecosystem preservation, and maritime safety on coastal water bodies is greatly aided by the MacroFloat dataset and standard. The study's findings aid in the creation of efficient remote sensing techniques for the detection of large floating objects in dynamic coastal environments. In another publication [6],[10], the dataset used was obtained by millimeter-wave radar which uses electro-magnetic waves in the frequency domain of 30 to 300 to obtain the synchronized images.

In the year 2022, identifying temporary water bodies from drone images at real-time using deep-learning techniques [2], was published in which they used data collected from UAV-Unmanned Aerial Vehicle. It gave us an idea on how to work with data extracted in a quite similar manner which is USV- Unmanned Surface Vehicle. We can see work done on the dataset which consists of infrared images[7]. This model suggests an advanced approach to detecting small objects in infrared images of water surfaces. This model helps in detecting any objects in water bodies even in bad weather. It integrates spatial feature weighting and a class balancing method to enhance the accuracy of detection in infrared images. The combination of these techniques aims to improve the model's sensitivity to small objects with low thermal signatures, common in water surfaces, ultimately enhancing detection performance in challenging infrared conditions.

The most recent publications i.e., in 2023 were quite insightful and helped us to know the working principle of YOLOv5. Soft-NMS-Enabled YOLOv5 with SIOU for Small Water Surface Floater Detection in UAV Captured Images[8], was one such work done using the UAV dataset and YOLOv5. A real-time object detection system called YOLOv5 creates a grid out of an image and forecasts the bounding boxes and class probabilities for each grid cell. The core architecture of YOLOv5 collects features from the input image and is often based on CSPNet or EfficientNet. In the post-processing stage, soft-NMS is included to improve the bounding box predictions. Soft-Maximum Suppression (Soft-NMS) prevents the removal of boxes with marginally lower confidence scores by giving overlapping bounding boxes lower scores than regular Non-Maximum Suppression (NMS). Subsequently, they used SIOU and Soft-NMS in the trained YOLOv5 model to detect floaters in photos taken by UAVs. The spatial distribution of nearby boxes is taken into account by the Soft-NMS method to improve bounding box predictions, while SIOU assesses the precision of the regions that are discovered. An analysis of Deep Learning Based Garbage Detection in Water Bodies

[1], was done using the most recent deep learning models available like DAE(Denoising Auto Encoding) and LSTM.

Based on the current publication works it is evident to tell that deep learning models can be of great help in detecting the waste in the water bodies. USV based dataset was used where we augmented it into different classes and worked using YOLO models to train and work on the problem. The accuracy of the earlier proposed models were quite low which is not suitable for our problem. So in order to improve accuracy and get better results we trained data using YOLO models with newer version activation functions such as Leaky ReLU, Hardswish, SiLU. We also added different loss functions. In addition to these we also included transformers such as SETR, DETR, C3tr etc., to help us get better results. We think that the suggested improvements can significantly increase the system's detection accuracy, which can have major uses in pollution management and environmental monitoring. To sum up, the goal of this research was to create a system that would use small item detection techniques to find small objects using the data captured by USV.

CHAPTER – 3

SYSTEM SPECIFICATIONS

3.1 Software requirements

- Operating system: YOLO can run on a variety of operating systems, including Linux, Windows, and macOS.
- Python: Python 3.7 or later is required.
- PyTorch: PyTorch 1.7 or later is required.
- CUDA and cuDNN (optional): CUDA and cuDNN are required if you want to use a GPU for training or inference. CUDA 10.2 or later and cuDNN 7.6.5 or later are recommended.
- OpenCV (optional): OpenCV is optional, but it is recommended for image preprocessing.

Additional recommendations:

- Use a high-performance storage device for your dataset. This will reduce the I/O overhead and improve the performance of the algorithm.
- Make sure that your system has enough memory available. Training and inference can be memory-intensive, especially for large datasets or high-resolution images.
- Use a GPU for training and inference if possible. This will significantly improve the performance of the algorithm.

3.2 Hardware requirements

- CPU: A modern CPU with at least 4 cores is recommended for basic object detection tasks. For more demanding tasks, such as real-time object detection or training large models, a GPU with at least 8GB of memory is recommended.
- Memory: A minimum of 8GB of RAM is recommended, but more is recommended for larger datasets or higher-resolution images.
- Storage: A fast SSD or NVMe drive is recommended for storing the dataset and model weights.

CHAPTER - 4

SYSTEM DESIGN

4.1 DATASET

ORCA-Uboat provides the FloW-IMG dataset, which we exploited in our investigation [3]. It is composed of 2,000.jpg images with more than 5,000 labeled floating garbage (small items) and annotations to help identify objects in the image.

Appropriate bounding box parameters and coordinates. Additionally, 200 real-time recorded films are given out without any commentary. The photographs are mostly taken in inland water bodies like lakes using a camera mounted on an unmanned surface vehicle. The dataset is made up of one class of floating water trash, which is plastic bottles. The resolution of the images is 1280*640 and, for the most part, 1280*720. The little objects under consideration are defined in a way that their area does not exceed 32 by 32 pixels.

Figure 3 provides some photos from the FloW-IMG dataset. The same shows many situations and difficulties that arise when detecting little objects.

Brightness or Contrast Faulty



Fig. 3A



Fig. 3B

Clear Visibility



Fig. 3C



Fig. 3D

Inappropriate Angle



Fig. 3E



Fig. 3F

No Bottle



Fig. 3G



Fig. 3H

Unclear and Blur Images



Fig. 3I



Fig. 3J

The dataset consists of images captured by USV(Unmanned Surface Vehicle). It has a wide set of images with various challenges. Figures 3A and 3B are examples of images that are a part of brightness and contrast faulty images. This set of images has either high brightness or low brightness or has some indeference in contrast which makes it difficult to detect the plastic bottles. Figures 3C and 3D are a part of set which consists of images named clearly visible in which bottles and cans can be easily detectable. Figures 3E and 3F are taken as examples from the set of images which has objects but in an inappropriate angle. 3G and 3H figuers are the examples of set which doesn't consist of a bottle or cans. Figure 3I and 3J are taken from the set of images

which are usually blur or unclear. This last set is also a bit difficult to identify the objects

4.2 LOW-LEVEL DESIGN WITH DESCRIPTION

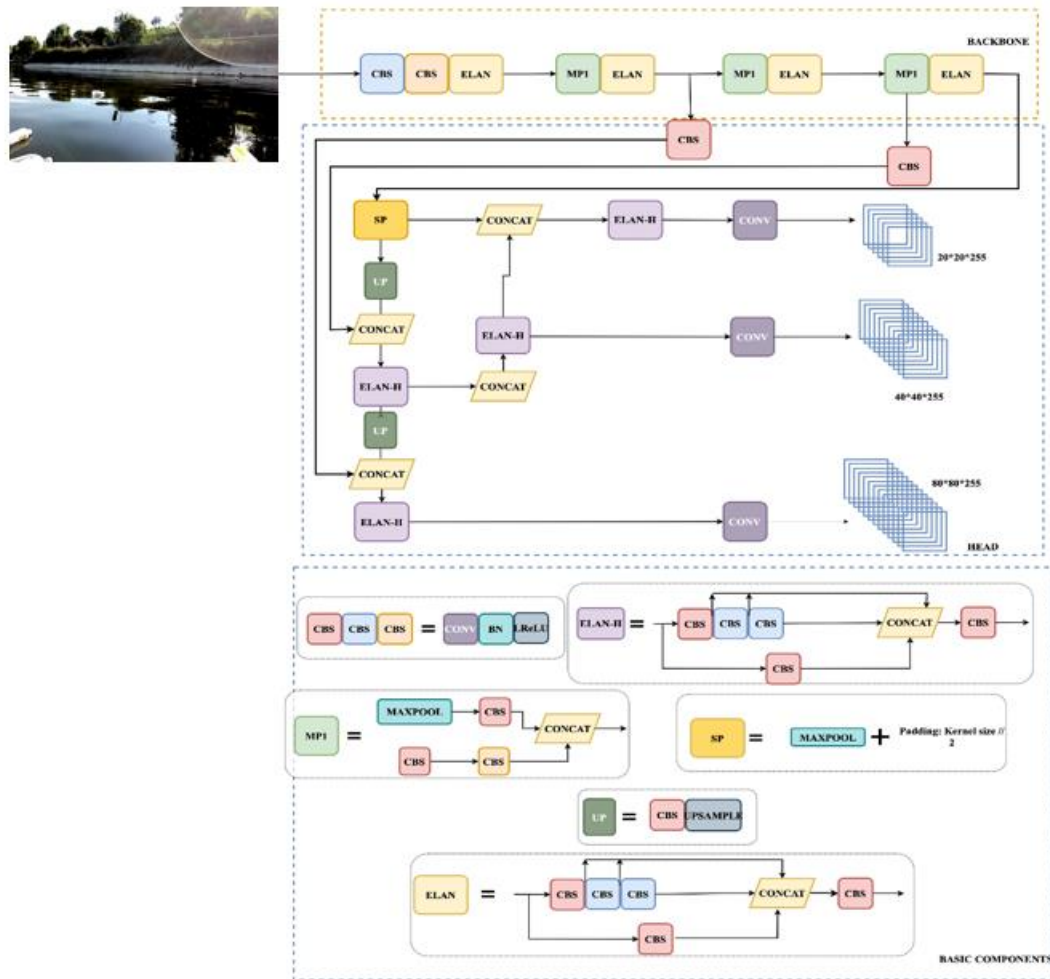


Fig 4. The architecture of yolov7 model

Three essential parts make up the YOLOv7 architecture, which is well-known for its skill at object detection: the head, neck, and backbone. Together, these components enable remarkably accurate feature extraction, bounding box prediction, and object classification. Let's examine each part in more detail:

1. Head:

- YOLOv7 utilizes multiple heads for different tasks:
 - Object detection head: This predicts bounding boxes and class probabilities for each detected object.

- Confidence head: This predicts the confidence score for each bounding box, indicating how certain the model is about the detection.
 - Classification head: This further refines the class prediction for each object.
- Focus mechanism: YOLOv7 employs a novel focus mechanism that dynamically adjusts object detection based on image content. This improves efficiency and accuracy, especially for small objects.

2. Neck:

- The neck acts as a bridge between the backbone and head: It processes and refines the feature maps extracted from the backbone, preparing them for the head's prediction tasks.
- YOLOv7 uses a Cross Stage Partial Connection (CSP) neck: This design reduces redundancy and improves information flow, leading to better performance compared to previous YOLO versions.

3. Backbone:

- The backbone is the foundation of the network: It extracts features from the input image at different scales and resolutions.
- YOLOv7 offers backbone flexibility: You can choose from various pre-trained backbones like CSPDarknet53, EfficientNet, and ConvNeXt, depending on your desired accuracy and inference speed trade-off.

4.3 HIGH LEVEL DESIGN WITH DESCRIPTION

The high-level architecture is shown in Figure 4 and consists of a number of modules, from detection to pre-processing and data gathering. Gathering information from the official ORCAUBOAT website, which offers pictures and PASCAL VOC XML annotations, is the first step. After that, a module parses the annotated data to produce the necessary YOLO standard format for use in YOLO. In addition, the pictures are cropped to match the 640×640 pixel input dimension of YOLOv7. The YOLOv7 model is then fed the training data and accompanying YOLO annotations; the parameters of this model are covered in the implementation part of this article. The dataset's YOLO-formatted test set is utilized for testing. Comet and ClearML are two visualization tools that are used to track the machine learning program's.

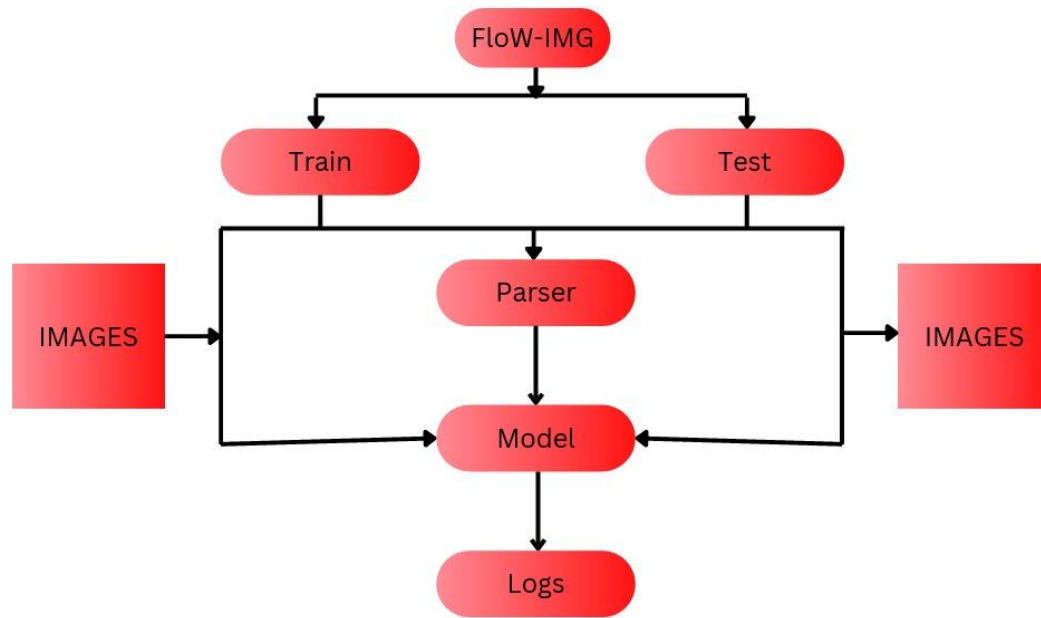


Fig 5 . Flow chart showing the high-level system design

4.4 MODEL ARCHITECTURE

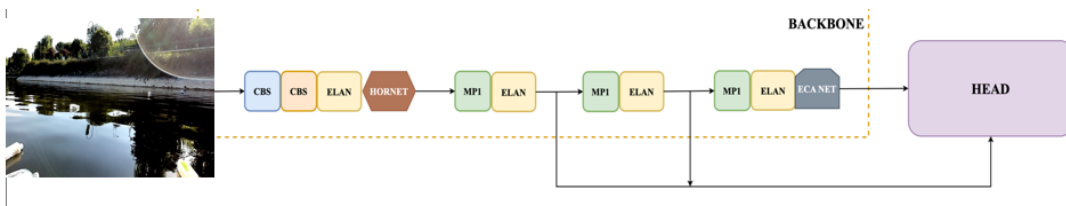


Fig 6 . Proposed model architecture after fine tuning the model.

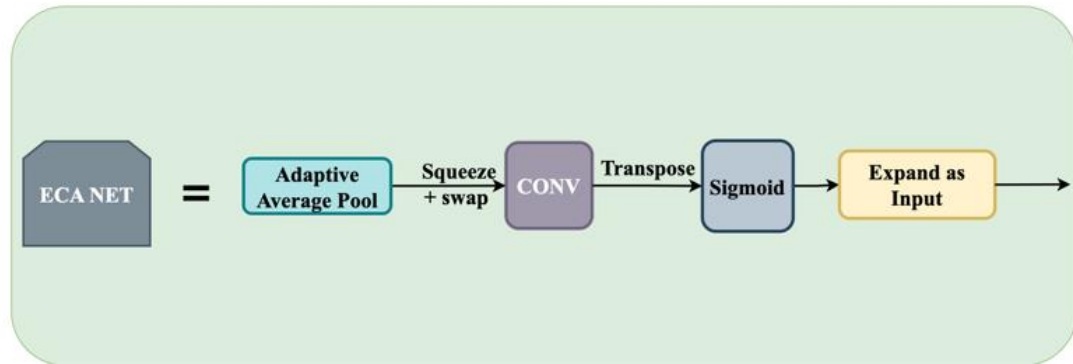


Fig 7 . ECA NET architecture

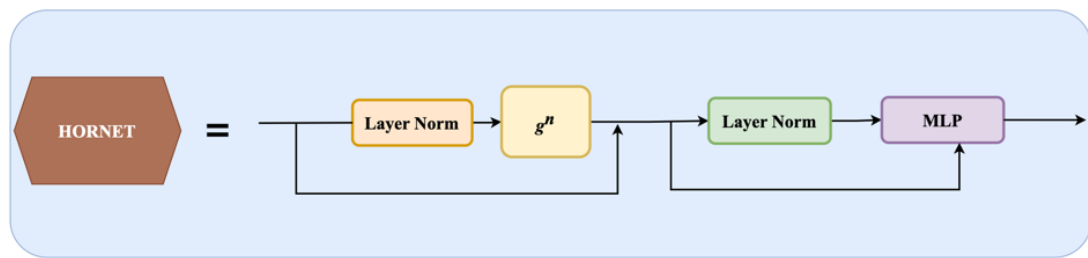


Fig 8 . HORNET architecture

Although the YOLOv7 architecture in its basic form has remarkable object detection capabilities, researchers are always looking for ways to improve its effectiveness. To increase feature extraction and accuracy, one such investigation is adding Hornet and ECA-Net to the backbone.

1. Hornet:

- Channel-wise transformers: Hornet replaces traditional convolutional layers with lightweight transformers operating on individual channels. This allows for better long-range feature interactions and dependencies, leading to higher-quality feature maps.
- Efficient computation: Hornet utilizes a simplified encoder-decoder structure and dynamic skip connections, resulting in lower computational cost compared to standard transformer architectures. This efficiency makes it suitable for real-time object detection on resource-constrained devices.

2. ECA-Net:

- Attention mechanism for channel enhancement: ECA-Net enhances feature maps by selectively amplifying informative channels and suppressing less informative ones. This improves the overall discriminability and representation of features.
- Lightweight design: ECA-Net employs a single-stage attention module with limited parameters, making it a lightweight addition to the backbone without significant performance overhead.

Combining HorNet and ECA-Net:

- Complementary effects: The long-range interactions of HorNet complement the channel-wise attention of ECA-Net, resulting in a synergistic effect that boosts feature representation and object detection accuracy.
- Backbone integration: Both modules can be seamlessly integrated into various YOLOv7 backbone stages, offering flexibility and adaptation based on specific needs.

Benefits of using HorNet and ECA-Net:

- Improved accuracy: Studies have shown that modified YOLOv7 architectures with HorNet and ECA-Net achieve higher object detection accuracy compared to the standard version, especially for smaller objects.
- Enhanced efficiency: HorNet's efficient design and ECA-Net's lightweight modules minimize the performance impact, making the modified architecture suitable for real-time applications.
- Flexibility and customization: The modular integration allows for adaptation to different backbone stages and specific task requirements.

The convolutional layers of the conventional YOLOv7 architecture use the ReLU activation function. Although successful, scientists are looking into different activations that can enhance the network's functionality. Here is a brief summary of the applications of SiLU, Leaky ReLU, and Parametric ReLU in modified YOLOv7 architectures:

1. Leaky ReLU:

- Addresses the "dying neurons" issue: Unlike ReLU, which sets negative values to zero, Leaky ReLU allows a small negative slope for non-zero activations. This prevents neurons from completely dropping out during training, potentially helping to maintain information flow and improve gradient propagation.
- Impact on performance: Studies have shown that Leaky ReLU can offer slight accuracy improvements compared to ReLU in certain object detection tasks, particularly for small objects. However, the optimal leakiness parameter needs careful tuning.

2. Parametric ReLU (PReLU):

- Dynamic activation values: PReLU introduces trainable learnable parameters for the negative slope of the activation function. This allows the network to adapt the slope for each neuron and layer, potentially leading to a better fit for the data.
- Increased complexity: Compared to Leaky ReLU, PReLU adds additional parameters and computational cost. Balancing performance gains with training overhead is crucial.

3. SiLU (Smoothly-clipped Exponential Linear Unit):

- Continuous differentiability: Unlike ReLU's sharp cutoff at zero, SiLU offers a smooth and continuous gradient, potentially improving optimization and convergence during training.
- Potential benefits: Similar to Leaky ReLU, SiLU may lead to improved accuracy and information flow through the network. However, its wider activation range compared to ReLU might require careful weight initialization and normalization strategies.

CHAPTER – 5

SYSTEM IMPLEMENTATION

Google Colaboratory was used to implement the project in Python 3.8.16. The sections that follow are separated into the following categories: Section 5.1 outlines the YOLO data preparation procedure; Section 5.2 discusses the model training and the parameters employed; Section 5.3 explains the evaluation measure; and, lastly, Section 5.4 provides an overview of the project's modular implementation.

5.1 DATA PREPROCESSING:

The Chinese USV manufacturer ORCA-Uboat's website provided the dataset utilised in this project. A sizable collection of floating bottle images with PASCAL Visual Object Classes XML annotations can be found in the FloW IMG dataset. The annotation format is intended to be standardised by the PASCAL VOC XML format. An example of the XML annotations may be seen in Figure 6. Each image has three RGB channels, so the depth tag is set to 3, and the size tag tells us how high the resolution of the original image is. Because this kind of item is the only one in the dataset, "bottles" is the class name that appears in the name tag. Consequently, rather than classifying or segmenting the objects, the goal is to detect and localise them. Since the goal is to localise items where the orientation may vary much, the pose tag is left undefined, meaning that the pose information is not very useful. The object's presence—whether completely or partially visible—is indicated by the truncated tag. All of the bounding box's required information, such as the height, width, and beginning and ending points in the X and Y axes, is contained in the bndbox tag.

The model chosen for small object detection is YOLOv5. The YOLO annotation standard input format for YOLO models is a text file with numerical values that constitute the bounding box. Nevertheless, the ORCA-Uboat website's data only has XML annotations in it. As a result, the first step is to transform the annotation data into a format that YOLO accepts. Parsing is a simple way to carry out this conversion. To read the XML file and parse it into the necessary text file with YOLO standard annotation, a straightforward Python function was created.

It's essential to comprehend what a YOLO annotation is in order to parse the XML annotations into that format. Every object in the text file is represented by a new line or row in the YOLO format. The object's class, the bounding box's height and breadth,

and the X and Y coordinates of its centre point are all contained in a single line. To fit inside the range of 0 to 1, all bounding box parameters are normalised.

5.2 TRAINING:

Preparing the YOLO standard annotations for the dataset of floating bottles in inland water bodies was the initial stage in our approach. In this approach, a little item was defined as one that covered an image with a maximum of 32 by 32 pixels. The YOLOv7 model was selected since it was pre-trained on the COCO dataset and has better accuracy than its predecessors. Using our dataset, the pre-trained model is refined through the application of transfer learning. 2000 photos and comments made up the dataset, which was divided into training and testing sets of three to two.

The goal was to identify and locate bottles, as the Flow dataset contained only one type of object: bottles. Pretrained weights acquired during training on the COCO dataset were used to initialise the model. The architecture was created so that it could be trained using 640 x 640 pixel input images. Table 1 lists the model's training parameters. As seen in Table 2, we applied a number of data augmentation methods, such as HSV, translation, scaling, horizontal flipping, and glaring, to further enhance the performance of our model. Hue, saturation, and value shifts were among the range of values that were applied to the HSV augmentation. Random applications were made to the translation, scaling, and horizontal flipping augmentations. To make the training more diverse, mosaic augmentation was applied over the entire set of training data. Google Colab Pro was used for the training process, and the GPU accelerator was used to speed up the process. The validation dataset was used to assess the model during training to make sure it wasn't overfitting to the training dataset.

Hyperparameter	Value
Learning Rate	0.01
Momentum	0.937
Weight Decay	0.0001 to 0.001.
Optimizer	Stochastic Gradient Descent
Epochs	100
Batch Size	16
Image Size	640
Dropout	0.3

Table 1: Hyperparameter used in model training

Augumentation	Values
Saturation	-25% and +25%
Brightness	-25% and +25%
Exposure	-25% and +25%

Table 2: Augmentation used in model training

5.3 METRICS:

A crucial component of assessing how well machine learning models work is their metrics. Metrics offer numerical evaluations for accuracy, precision, recall, and other performance metrics of a model. Mean average precision (mAP) is one of the metrics for object detection models that is most frequently employed. The precision and recall of a model's item detection inside an image are measured by its mean average precision, or mAP. The precision values at different recall levels are averaged to get the measure. The 0.5:0.95 mAP is a widely used variant of mAP that takes precision values over a range of IoU (Intersection over Union) thresholds, from 0.5 to 0.95. This metric is especially helpful for assessing object identification models since it provides a more thorough evaluation of the model's performance by accounting for a variety of object sizes, shapes, and orientations.

5.4 MODULES:

The suggested system has a distributed architecture in which several modules interact with one another to perform as a unit. The main folder has a Python file called "train.py" that is used to train the model throughout the training phase. The "yolov5x.yaml" file contains the YOLO model, which is loaded by the "yolo.py" file in the "models" folder to start training. Training starts when "DataLoader.py" in the "utils" folder imports the necessary training data into the model. After every iteration, the "train.py" programme calls "val.py" to confirm that the model has learned. Recall, precision, mAP, mmAP, and losses—which are computed by "loss.py"—are computed by the "metrics.py" programme located in the "utils" folder. These metrics are logged in "utils" using "loggers.py" and provided to "train.py". Modifying the Comet ML platform to produce a log file and enable visualisation with the required details, such as the weights of the model.

The previously saved weights from the training phase are loaded using "detect.py" in the main folder during the detection or inference phase. "Detect.py" uses the trained weights to detect and localise objects, while "DataLoader.py" loads the photos for detection, same like in the training phase. The pictures and development are kept in logs for a future examination.

CHAPTER - 6

SYSTEM TESTING

The Flow-Img dataset contains 2000 images, in which 60% of the images i.e., 1200 images are used for training and remaining 40% i.e., 800 images are used for validation and testing. As there were many challenges faced with the images in the dataset and classified as Clear Visibility, Inappropriate Angle, Glared Images, Unclear or blur images, Diminished. As the model should also be able to detect while this type of images is given for testing. So, data augmentation is done with the dataset by adding Saturation: Between -50% and +50%, Brightness: Between -25% and +25%, Exposure: Between -25% and +25% and add those images to the training set. After data augmentation, the training set consists of 3600 images.

On 800 photos from the FloW-IMG dataset's test set, the suggested system is put to the test. The loss function of YOLOv7 is the sum of three losses: objectness loss, location loss, and box loss, all of which are covered in Chapter 4. Since there is just one object class in our situation, the class loss will be 0. Figures 9 show, respectively, the plots of the training losses and validation losses. These figures suggest that the intended reduction in training and validation losses has occurred. Three measures are examined, which are recall, precision, and mean average precision scores (mAP 0.5 and mAP 0.5:0.95). As stated in, graphs for the same are produced and shown using Comet and ClearML.

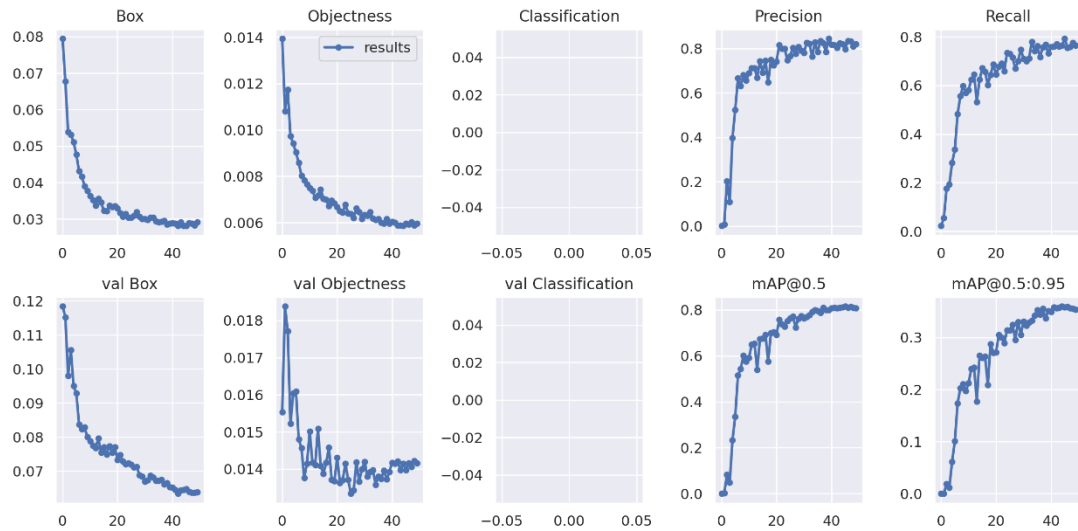


Fig 9 . Performance metrics of the proposed model.

From the graphs presented in Fig 6, It is observed that the training box loss, class loss and dfl loss are decreasing which depicts the model is training well. If the loss is decreasing during the training of a model, it generally indicates that the model is learning and improving its ability to make predictions on the training data. Loss is a measure of how well the model's predictions match the actual target values in the training set. Additionally, it is observed from Fig 6 that the validation loss is also decreasing gradually. If the validation loss is decreasing during the training of a model, it is a positive indication that the model is learning to generalize well to new, unseen data. This suggests that the model is avoiding overfitting, improving its applicability to real-world scenarios, and may warrant considerations for hyperparameter tuning and model selection. Monitoring validation loss is crucial for ensuring the model's ability to perform well on data it has not encountered during training. It is also observed that the precision and recall are increasing which indicates that the model is make a decent prediction after training. If the precision and recall graph is increasing, it signifies positive advancements in the model's performance. The model is improving its ability to make accurate positive predictions (precision) while capturing a larger portion of actual positive instances (recall). This suggests a balanced trade-off and effective decision-making. The increasing graph may result from model refinements, threshold adjustments, or iterative changes, reflecting the model's reliability and positive impact in various applications.

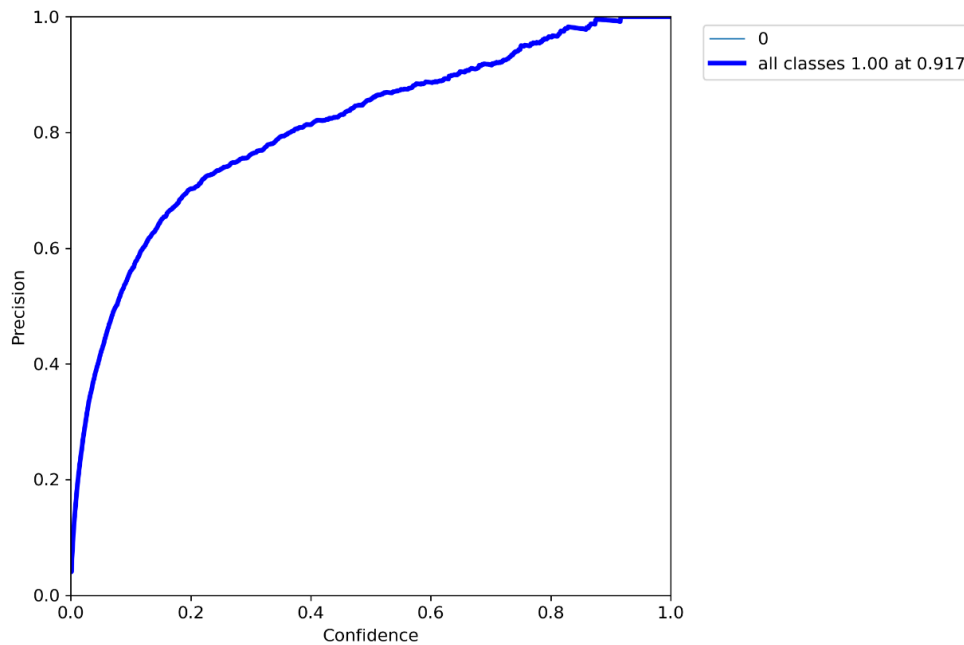


Fig 10. Precision Curve of the proposed model

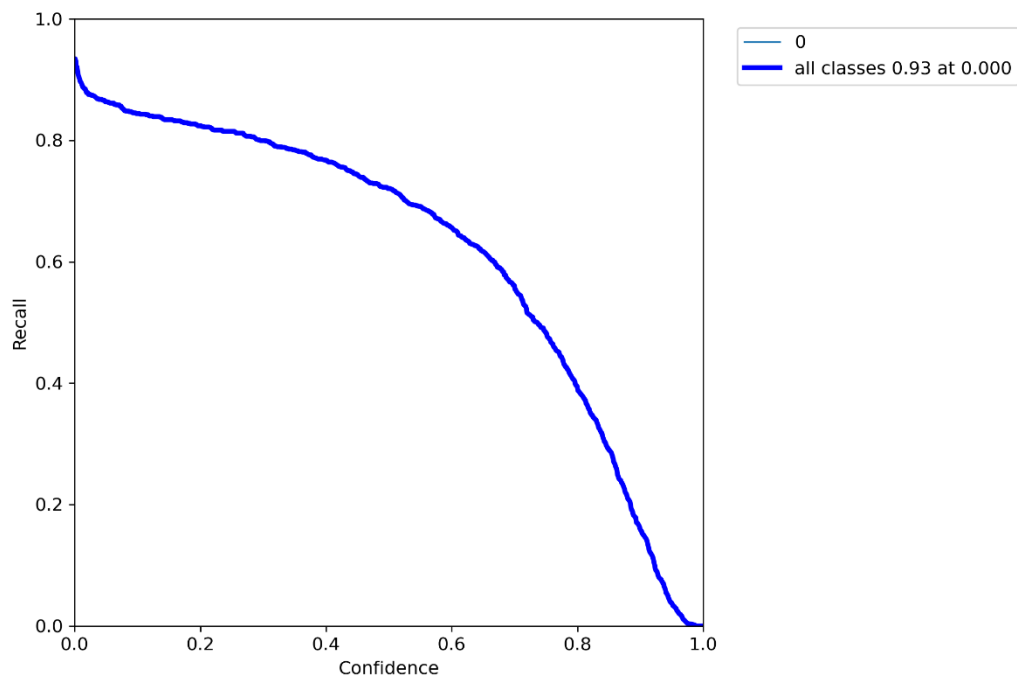


Fig 11. Recall curve of the proposed model.

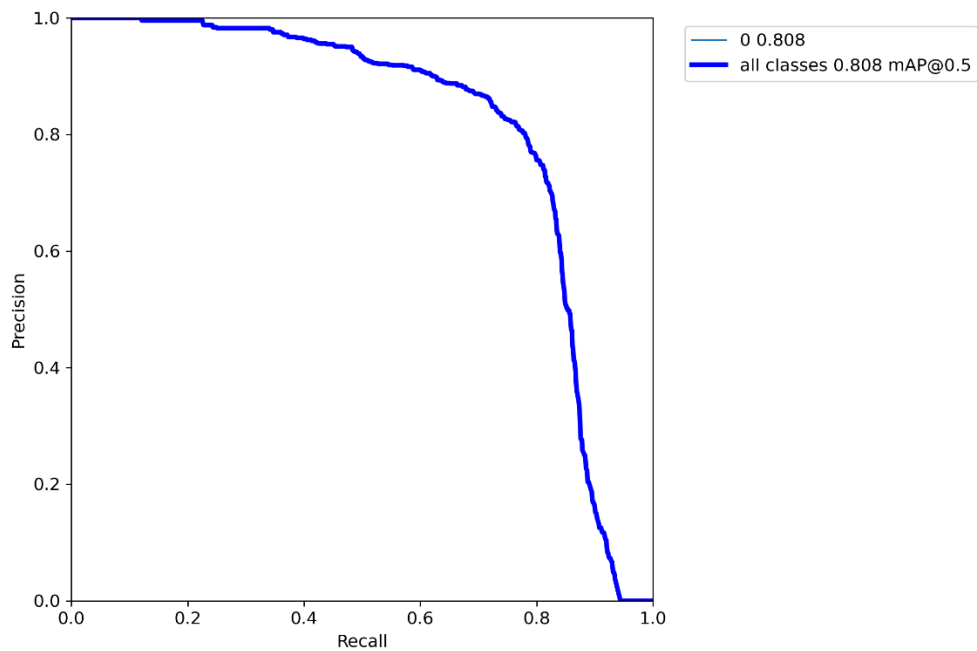


Fig 12. Precision and Recall curve of the proposed model.

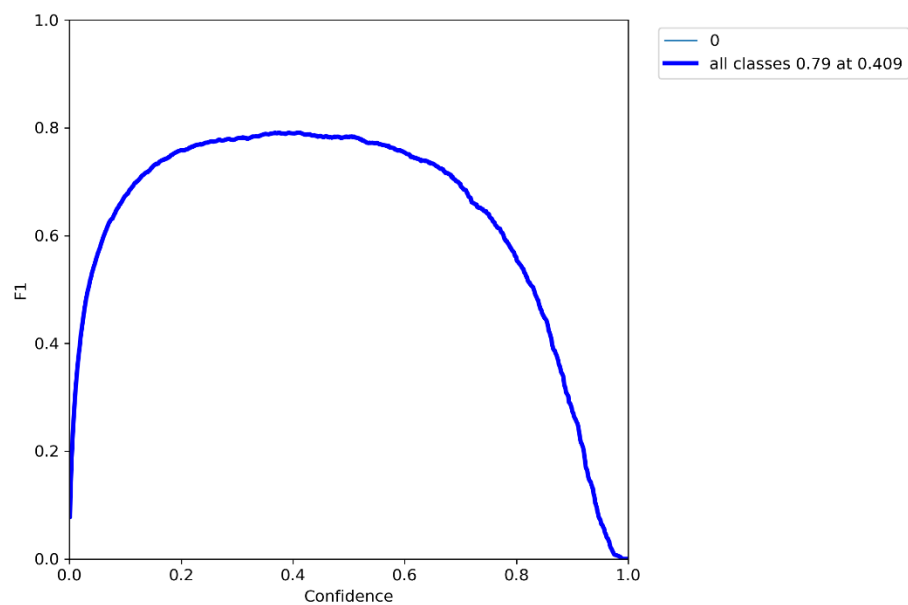


Fig 13. F1 Score curve of the proposed model.

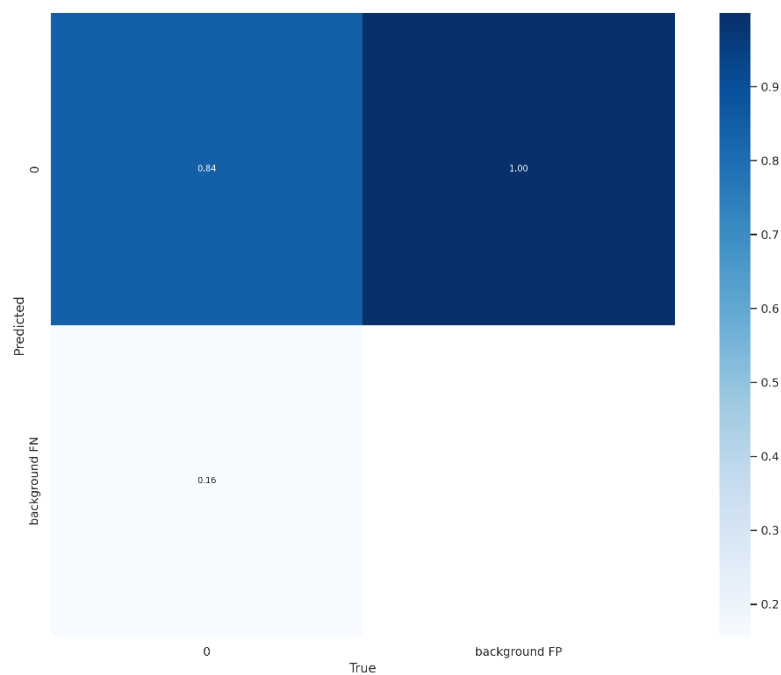


Fig 14. Confusion matrix of the proposed model.

Additional experiments have been conducted to evaluate the suggested model's performance using a few pictures taken from a Bangalore lake. Make broad generalizations. The performance on these images has been examined in light of the particular difficulties covered in Chapter 4.

CHAPTER – 7

RESULTS AND ANALYSIS

With regard to floating bottle detection, our proposed model obtained a mAP score of 40.2%, representing a 3.4% improvement in accuracy over the baselines suggested in [3]. Table 3 compares the performance of the proposed model with the baselines that are provided for the FloW IMG dataset.

Models	Precision/%	Recall/%	mAP 0.5/%	mAP 0.5:0.95/%
Yolo V8	78.0	68.0	77.4	36.8
Yolo NAS	16.1	87.4	76.1	35.6
Yolo V7	21.6	19.6	11.3	3.24
Yolo V5s	46.4	34.6	34.4	19.0
Yolo V4	36.0	48.6	42.1	25.7

Table 3. Results after training with different models.

Models	Precision/%	Recall/%	mAP 0.5/%	mAP 0.5:0.95/%
yolov7-tiny architecture [Leaky ReLu]	81.7	81.9	84.3	38.8
yolov7-tiny- PRelu	83.9	81.2	84.7	39.5
yolov7-tiny- ecanet-NO- SPPCSPC	82.0	77.4	82.2	36.7
yolov7-tiny- hornet2b-pipe	83.9	81.2	84.7	39.5
yolov7-tiny- hornet- ecanet-pipe	84.2	82.3	85.8	40.2

yolov7-nano- hornet-ecanet- pipe	75.5	52.3	58.7	21.8
--	------	------	------	------

Table 4. Results after training with different yolov7 models.

The following parts can be used to further separate and examine the results section: The test dataset prediction is examined in Section 7.1, and the model's performance is compared in Section 7.2 based on the differences in activation functions.

7.1 Analyzing model predictions on test set



Fig. 15A



Fig. 15B



Fig. 15I



Fig. 15J



Fig. 15K



Fig. 15L



Fig. 15M



Fig. 15N



Fig. 15O



Fig. 15P



Fig. 15Q



Fig. 15R



Fig 15S,T. Prediction of test images made by the proposed model

Similar to the photos in the training set, the images shown in Figure 15 are indicative of the test set. High light reflections on the water's surface are seen in Images 15.a and 15.e, and the model shows precise predictions that hold up well to surface reflections. However, due to extra considerations like the target object's tiny size and decreased clarity in image 15.a, the model's confidence score is quite low. Images 15.b, 15.g, 15.h, and 15.k show several instances of bottles at various levels of complexity. The model correctly detects every bottle in pictures 15.b and 15.h, when they are all clearly visible and spaced apart. In image 15.g, the bottles exhibit some degree of overlap, but the model is able to distinguish between five different bottles, but the confidence scores are somewhat reduced.

Only a small portion of the bottles in image 15.k can be correctly identified since, despite their clear placement, their size has greatly decreased. Pictures 15.c and 15.f show how well the model performs in a situation involving bottle camouflage, where the model fails to identify the bottles. This is because, even with the naked eye, it is difficult to detect the bottles because they are well-hidden in the background. Despite the target bottles' minuscule size, Image 15.d shows the proper identification of them. But the model also predicts some incorrect things, which is understandable given the image's dearth of information that would otherwise aid in differentiating bottles from other objects. Image 15.j, where occlusion causes the model to get inadequate context from the image. There are instances of over-lit target objects in images 15.d and 15.i. On the other hand, image 15.i's significantly larger bottle gives the model a clearer context, which leads to a correct detection.

The model is resistant to strong light reflections on the water's surface and can reliably identify bottles of various complexities while they are still clearly visible, according to the examination of the model's performance on the photos shown in Figure 15. But in situations when the bottles are obscured or camouflaged, the model has trouble. In addition, occlusion, insufficient data, and the target object's size and clarity all have an impact on the model's performance.

7.2 Experimental analysis of PReLU, Leaky ReLU and SiLU activations on model performance

Selecting the appropriate activation function is a crucial step in developing a successful deep learning model. In the suggested model, we compare three activation functions in this work: PReLU, Leaky ReLU and SiLU. It has been demonstrated that the non-linear activation function SiLU performs more accurately than more conventional activation functions like ReLU. Equation 1 provides the equation for Swish activation. By adding

a tiny slope for negative input values, the Leaky ReLU activation function, a variation of the ReLU activation function, circumvents the "dying ReLU" issue. Equation 2 provides the equation for leaky ReLU activation. Conversely, PReLU is a computationally effective non-linear activation function that was recently suggested as a substitute for Swish. The Equation 3 provides the PReLU activation equation. Based on our studies, as seen in Table 4, the PReLU activation function performed faster and more accurately than the other two activation functions.

$$f(x)=\max(0.01*x, x) \quad \text{Equation 1}$$

$$f(x)=\max(0,y)+a*\min(0,y) \quad \text{Equation 2}$$

$$f(x)=x*\sigma(x)$$

$$\sigma(x) = 1/1+e^{-x} \quad \text{Equation 3}$$

Models	Precision/%	Recall/%	mAP 0.5/%	mAP 0.5:0.95/%
yolov7-tiny architecture [Leaky ReLu]	81.7	81.9	84.3	38.8
yolov7-tiny- PRelu	83.9	81.2	84.7	39.5
yolov7-tiny- SiLU	82.6	81.1	84.1	37.9

Table 5. Impact of Different Activation Functions on Model Performance

CHAPTER – 8

CONCLUSION AND FUTURE SCOPE

We evaluate several conventional vision-based object detection algorithms and radar detectors on our dataset. The results of the experiment show that in order to meet the requirements of real-world applications, the robustness of the model for vision-based object detection must be strengthened. We anticipate that the cleanup of wastes in water regions will receive more attention, and that detection of floating wastes may become more accurate. In some unclean areas, we will collect more types of floating trash for more categories in the future to encourage the use of autonomous cleaning in inland waterways. As we have already trained with the different versions of the yolo with small, medium and large size object detections and compared with the mAP's of each model. The future scope of the project is to modify the architecture of the yolo versions and add attention mechanism, bounding box algorithm and the modify the backbone architecture so that the mAP of the proposed model will give the satisfactory results.

REFERENCES

1. R. Carmo, J. Mifdal and M. Rußwurm, "Detecting Macro Floating Objects on Coastal Water Bodies using Sentinel-2 Data," OCEANS 2021: San Diego – Porto, San Diego, CA, USA, 2021, pp. 1-7, doi: 10.23919/OCEANS44145.2021.9705668.
2. Harsh Panwar, P.K. Gupta, Mohammad Khubeb Siddiqui, Ruben MoralesMenendez, Prakhar Bhardwaj, Sudhansh Sharma, Iqbal H. Sarker, AquaVision: Automating the detection of waste in water bodies using deep transfer learning, Case Studies in Chemical and Environmental Engineering.
3. V. Vijayanti, A. Kar, N. K. N. S. S and S. S. S, "Analysis of Deep Learning Based Garbage Detection in Water Bodies," 2023 5th International Conference on Inventive Research in Computing Applications (ICIRCA), Coimbatore, India, 2023, pp. 1-8, doi: 10.1109/ICIRCA57980.2023.10220828.
4. A. Mahanti, A. Banerjee and K. Srinivasan, "Enabling Detection of Plastic Debris on Water Surface Using Wireless Signals," in IEEE Sensors Letters, vol. 5, no. 7, pp. 1-4, July 2021, Art no. 3500904, doi: 10.1109/LSSENS.2021.3090093.
5. H. M. Truong and M. Clavel, "Identifying temporary water bodies from drone images at real-time using deep-learning techniques," 2022 International Conference on Advanced Computing and Analytics (ACOMPA), Ho Chi Minh City, Vietnam, 2022, pp. 12-19, doi: 10.1109/ACOMPA57018.2022.00009.
6. Y. Cheng, H. Xu and Y. Liu, "Robust Small Object Detection on the Water Surface through Fusion of Camera and Millimeter Wave Radar," 2021 IEEE/CVF International Conference on Computer Vision (ICCV), Montreal, QC, Canada, 2021, pp. 15243-15252, doi: 10.1109/ICCV48922.2021.01498.
7. Hui, Tian & Xu, YueLei & Li, HuaFeng & Zhou, Qing & Rasol, Jarhinbek. (2023). Water-surface infrared small object detection based on spatial feature weighting and class balancing method. IET Image Processing. 17. n/a-n/a. 10.1049/ipr2.12851.
8. Chen, Fuxun & Zhang, Lanxin & Kang, Siyu & Chen, Lutong & Dong, Honghong & Li, Dan & Wu, Xiaozhu. (2023). Soft-NMS-Enabled YOLOv5 with SIOU for Small Water Surface Floater Detection in UAV-Captured Images. Sustainability. 15. 10751. 10.3390/su151410751.

9. Zhou Z, Sun J, Yu J, Liu K, Duan J, Chen L, Chen CLP. An Image-Based Benchmark Dataset and a Novel Object Detector for Water Surface Object Detection. *Front Neurorobot.* 2021 Sep 24;15:723336. doi: 10.3389/fnbot.2021.723336. PMID: 34630064; PMCID: PMC8497741.
10. Y. Cheng et al., "FloW: A Dataset and Benchmark for Floating Waste Detection in Inland Waters," 2021 IEEE/CVF International Conference on Computer Vision (ICCV), Montreal, QC, Canada, 2021, pp. 10933-10942, doi: 10.1109/ICCV48922.2021.01077.
11. Litchfield SG, Schulz KG, Kelaher BP. The influence of plastic pollution and ocean change on detrital decomposition. *Marine pollution bulletin.* 2020 Sep 1; 158:111354.
12. Lebreton L, Slat B, Ferrari F, Sainte-Rose B, Aitken J, Marthouse R, Hajbane S, Cunsolo S, Schwarz A, Levivier A, Noble K. Evidence that the Great Pacific Garbage Patch is rapidly accumulating plastic. *Scientific reports.* 2018 Mar 22;8(1):1-5.
13. Cheng, Y., Zhu, J., Jiang, M., Fu, J., Pang, C., Wang, P., Sankaran, K., Onabola, O., Liu, Y., Liu, D. and Bengio, Y., 2021. Flow: A dataset and benchmark for floating waste detection in inland waters. In *Proceedings of the IEEE/CVF International Conference on Computer Vision* (pp. 10953-10962).
14. Kong S, Tian M, Qiu C, Wu Z, Yu J. IWSCR: An intelligent water surface cleaner robot for collecting floating garbage. *IEEE Transactions on Systems, Man, and Cybernetics: Systems.* 2020 Jan 9;51(10):6358-68.
15. Wu Q, Liu Y, Li Q, Jin S, Li F. The application of deep learning in computer vision. In *2017 Chinese Automation Congress (CAC) 2017 Oct 20* (pp. 6522-6527). IEEE.
16. Zhan, W., Sun, C., Wang, M., She, J., Zhang, Y., Zhang, Z. and Sun, Y., 2022. An improved Yolov5 real-time detection method for small objects captured by UAV. *Soft Computing*, 26(1), pp.361-373.
17. Lian, J., Yin, Y., Li, L., Wang, Z. and Zhou, Y., 2021. Small object detection in traffic scenes based on attention feature fusion. *Sensors*, 21(9), p.3031.
18. Liu, Y., Sun, P., Wergeles, N. and Shang, Y., 2021. A survey and performance evaluation of deep learning methods for small object detection. *Expert Systems with Applications*, 172, p.114602.
19. Zailan, N.A., Khairuddin, A.S.M., Khairuddin, U. and Taguchi, A., 2021, June. YOLO-based Network Fusion for Riverine Floating Debris Monitoring

- System. In 2021 International Conference on Electrical, Communication, and Computer Engineering (ICECCE) (pp. 1-5). IEEE.
20. Zhang, L., Zhang, Y., Zhang, Z., Shen, J. and Wang, H., 2019. Real-time water surface object detection based on improved faster R-CNN. *Sensors*, 19(16), p.3523.
 21. Tharani, M., Amin, A.W., Maaz, M. and Taj, M., 2020. Attention neural network for trash detection on water channels. *arXiv preprint arXiv:2007.04639*.
 22. Panwar, H., Gupta, P.K., Siddiqui, M.K., Morales-Menendez, R., Bhardwaj, P., Sharma, S. and Sarker, I.H., 2020. AquaVision: Automating the detection of waste in water bodies using deep transfer learning. *Case Studies in Chemical and Environmental Engineering*, 2, p.100026.
 23. Chang, H.C., Hsu, Y.L., Hung, S.S., Ou, G.R., Wu, J.R., and Hsu, C. (2021). "Autonomous water quality monitoring and water surface cleaning for unmanned surface vehicle." *Sensors*, 21(4), p.1102.
 24. Phirke, S., Patel, A., and Jani, J. (2021). "Design of an autonomous water cleaning bot." *Materials Today: Proceedings*, 46, pp.8742-8747.
 25. N. A. Zailan, A. S. Mohd Khairuddin, U. Khairuddin, and A. Taguchi (2021). "YOLO-based Network Fusion for Riverine Floating Debris Monitoring System." 2021 International Conference on Electrical, Communication, and Computer Engineering (ICECCE), Kuala Lumpur, Malaysia, pp. 1-5, doi: 10.1109/ICECCE52056.2021.9514096.
 26. Lin, F., Hou, T., Jin, Q., and You, A. (2021). "Improved YOLO-based detection algorithm for floating debris in waterway." *Entropy*, 23(9), p.1111.
 27. Topouzelis, K., Papageorgiou, D., Suaria, G., and Aliani, S. (2021). "Floating marine litter detection algorithms and techniques using optical remote sensing data: A review." *Marine Pollution Bulletin*, 170, p.112675.
 28. Carmo, R., Mifdal, J., and Rußwurm, M. (2021, September). "Detecting macro floating objects on coastal water bodies using Sentinel-2 data." *OCEANS 2021: San Diego–Porto*, pp. 1-7.
 29. Cheng, Y., Xu, H., and Liu, Y. (2021). "Robust small object detection on the water surface through fusion of camera and millimeter wave radar." *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 15263-15272.

30. Zailan, N.A., Azizan, M.M., Hasikin, K., Khairuddin, A.S.M., and Khairuddin, U. (2022). "An automated solid waste detection using the optimized YOLO model for riverine management." *Frontiers in Public Health*, 10.
31. Gómez, À.S., Scandolo, L., and Eisemann, E. (2022). "A learning approach for river debris detection." *International Journal of Applied Earth Observation and Geoinformation*, 107, p.102682.
32. Tong, K., Wu, Y., and Zhou, F. (2020). "Recent advances in small object detection based on deep learning: A review." *Image and Vision Computing*, 97, p.103910.
33. Liu, Y., Sun, P., Wergeles, N., and Shang, Y. (2021). "A survey and performance evaluation of deep learning methods for small object detection." *Expert Systems with Applications*, 172, p.114602.
34. Li, J., Liang, X., Wei, Y., Xu, T., Feng, J., and Yan, S. (2017). "Perceptual generative adversarial networks for small object detection." *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 1222-1230.
35. He, K., Gkioxari, G., Dollár, P., and Girshick, R. (2017). "Mask R-CNN." *Proceedings of the IEEE international conference on computer vision*, pp. 2961-2969.
36. Hu, G.X., Yang, Z., Hu, L., Huang, L. and Han, J.M., 2018. Small object detection with multiscale features. *International Journal of Digital Multimedia Broadcasting*, 2018.
37. Cao, C., Wang, B., Zhang, W., Zeng, X., Yan, X., Feng, Z., Liu, Y. and Wu, Z., 2019. An improved faster R-CNN for small object detection. *Ieee Access*, 7, pp.106838-106846.
38. Nguyen, N.D., Do, T., Ngo, T.D. and Le, D.D., 2020. An evaluation of deep learning methods for small object detection. *Journal of electrical and computer engineering*, 2020, pp.1-18.
39. Lim, J.S., Astrid, M., Yoon, H.J. and Lee, S.I., 2021, April. Small object detection using context and attention. In *2021 International Conference on Artificial Intelligence in Information and Communication (ICAIIIC)* (pp. 181-186). IEEE.
40. Lian, J., Yin, Y., Li, L., Wang, Z. and Zhou, Y., 2021. Small object detection in traffic scenes based on attention feature fusion. *Sensors*, 21(9), p.3031.

41. Benjumea, A., Teeti, I., Cuzzolin, F. and Bradley, A., 2021. YOLO-Z: Improving small object detection in YOLOv5 for autonomous vehicles. arXiv preprint arXiv:2112.11798.
42. Wu, Q., Feng, D., Cao, C., Zeng, X., Feng, Z., Wu, J. and Huang, Z., 2021. Improved mask R-CNN for aircraft detection in remote sensing images. *Sensors*, 21(8), p.2618.
43. Wang, X., Zhao, Q., Jiang, P., Zheng, Y., Yuan, L. and Yuan, P., 2022. LDS-YOLO: A lightweight small object detection method for dead trees from shelter forest. *Computers and Electronics in Agriculture*, 198, p.107035.
44. Liu, H., Sun, F., Gu, J. and Deng, L., 2022. Sf-yolov5: A lightweight small object detection algorithm based on improved feature fusion mode. *Sensors*, 22(15), p.5817.
45. Wang, N., Qian, T., Yang, J., Li, L., Zhang, Y., Zheng, X., Xu, Y., Zhao, H. and Zhao, J., 2022. An Enhanced YOLOv5 Model for Greenhouse Cucumber Fruit Recognition Based on Color Space Features. *Agriculture*, 12(10), p.1556.
46. Bosquet, B., Cores, D., Seidenari, L., Brea, V.M., Mucientes, M. and Del Bimbo, A., 2023. A full data augmentation pipeline for small object detection based on generative adversarial networks. *Pattern Recognition*, 133, p.108998.
47. Li, S., Liu, S., Cai, Z., Liu, Y., Chen, G. and Tu, G., 2023. TC-YOLOv5: rapid detection of floating debris on raspberry Pi 4B. *Journal of Real-Time Image Processing*, 20(2), p.17.
48. Redmon, J., Divvala, S., Girshick, R. and Farhadi, A., 2016. You only look once: Unified, real-time object detection. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 779-788).
49. Glenn Jocher, 2020. YOLOv5 by Ultralytics (version 7.0). <https://github.com/ultralytics/yolov5>.
50. Li, P., Wang, H. and Huang, X., 2022, July. Wheat straw target detection algorithm based on improved YOLOv5. In *2022 3rd International Conference on Information Science, Parallel and Distributed Systems (ISPDS)* (pp. 380-385).